

Alex O'Brien

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in a03541

🏠 SYDNEY, NSW

Systems programmer.

Education

2020–2022

Bachelor of Science, The University of Melbourne

Majored in Computing and Software Systems. Graduated with *First-Class Honours*. Relevant subjects include *Algorithms and Data Structures*, *Computer Systems*, *Probability*, and *Statistics*.

Skills

Technical

- Fundamentals** Deep understanding of hardware and software architecture — all the way from CPU architecture to operating systems implementation.
- Programming** Experience in a variety of languages and paradigms, including object-oriented and functional practices. Most fluent with C++ and Rust, with working knowledge of Python and Java, plus others. Very capable of learning on the fly and picking up new technologies quickly.
- Security** Understanding of security best-practices and fundamentals with common services (e.g., Apache and nginx), with experience in implementation of real systems.

Communication

- Written** Strong written communication skills both in an informative and argumentative style.

Experience

References available upon request.

Aug. 2022 –

Software Engineer, IMC Trading

LOW-LATENCY

ALGORITHM DEVELOPMENT

Worked across a variety of systems at various levels of a mature low-latency trading stack:

- ▷ Joined working on low-latency execution and connectivity on a growing desk.
- ▷ Transitioned to baseval algorithm development and research support.
- ▷ Now building a new execution strategy framework, designed from scratch for D1.

Mar. 2022 – Jun. 2022

Tutor, The University of Melbourne

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Taught labs and tutorials, marked exams and projects for Computer Systems (COMP30023).

Nov. 2021 – Feb. 2022

Software Engineer Intern, IMC Trading

CONCURRENCY

NETWORKING

PERFORMANCE

Worked on microsecond-sensitive concurrent infrastructure, delivering a project which provides monitoring of a high-performance IPC system with near-zero overhead.

Mar. 2021 – Dec. 2021

Debating Tutor, John Monash Science School

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Co-instructed the school’s VCE debate team.

Jun. 2016 – Aug. 2017

IT Technician, University Prep

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General IT support and infrastructure maintenance.

Personal Projects

Names are hyperlinked to the relevant repository.

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Short Circuit, Web Server

SERVICES

NETWORKING

HTTP

A high-performance web server for Linux using the new `io_uring` asynchronous I/O interface.

- ▷ Unix network programming.
- ▷ HTTP parsing and implementation.
- ▷ Performance profiling and optimization.

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liba3, Utility Library

C

LIBRARY DESIGN

DATA STRUCTURES

A lightweight utility library in C implementing useful features and data structures absent from the standard library.

- ▷ More ergonomic strings.
- ▷ Type-generic hash table, cache, list, and priority queue.
- ▷ Growable buffer.
- ▷ Object pool allocator.

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4cc, C Compiler

COMPILERS

PARSING

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A C compiler, in C. The ultimate goal is to self-host, but it can already compile a couple “real” programs (see the test suite).

Syzygy, Kernel

OPERATING SYSTEMS

COMPUTER ARCHITECTURE

A kernel implemented in Rust, currently featuring physical and virtual memory management, an `initramfs`, and interrupt handling. Presently working on multitasking.

- ▷ Kernel and bare-metal programming.
- ▷ `no-std` Rust programming.
- ▷ `x86_64` architecture and assembly.